IPsec group 3 GHS search evidence

This study archives the Python investigation completed outside this repository on 2026-10-23 through 2026-10-25. The saved checkpoint reports 24,492 Frobenius orbits screened and zero low-genus candidates in the implemented tests. The entire isogeny class has not been enumerated. This archive does not establish resistance to arbitrary Weil descent or a discrete-log attack.

The executable Python sources are historical snapshots, preserved under the historical-evidence exception in AGENTS.md. They are not a new supported research execution path. No new research computation was run for this import. Native continuation of the checkpointed degree-3/17/31 search remains open.

Question and exact input

The requested search targets the whole base-field isogeny class of RFC 2409 Oakley group 3, using low-genus GHS screens to find candidate representatives. The working field and model are

```
K = F2[z]/(z^155 + z^62 + 1)
E: y^2 + x*y = x^3 + b, a=0, b=0x7338f
N = 4567192616659071619386556591434463519676923731 6
r = 3805993847215893016155463826195386266397436443
N = 12*r
```

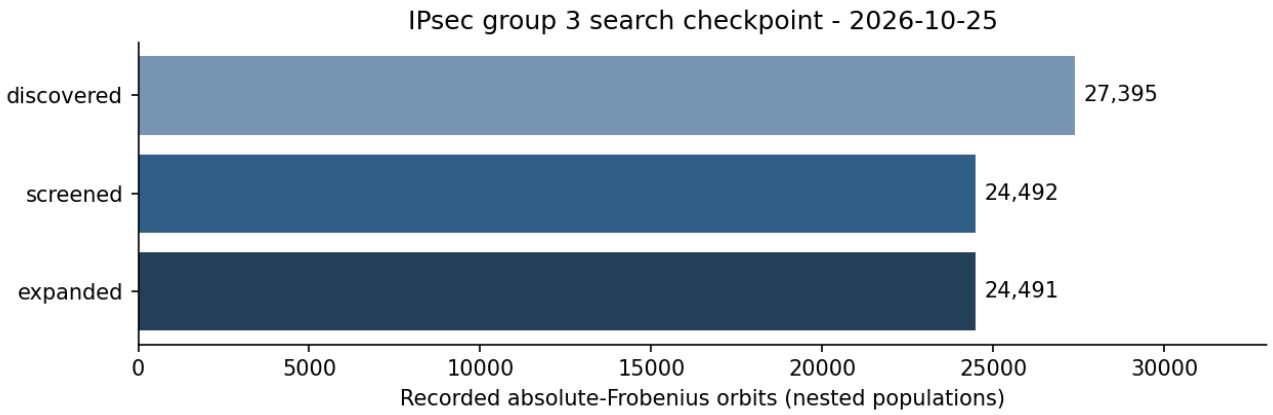
The intended subgroup is the order-r subgroup of the RFC model. This search does not select a generator or produce cross-repository EC1/ICV1 map-route records. Modular j-correspondences must not be read as certified subgroup transport. The historical degree-three maps have separate identity and sample-point checks; degrees 17 and 31 have modular-root witnesses only.

Recorded findings

Quantity	Recorded result	Evidence and scope
Seed direct GHS to F32	m=26; genus 33,554,432	historical_python/results.json
JMS comparison curve direct GHS to F32	m=6; genus 31	Same file; comparison curve, not the IPsec class
Seed Hess two-factor necessary genus bound over F32	992	historical_python/bound_certificate.json; 46 lower support pairs excluded
Search checkpoint, 2026-10-25	24,492 screened orbits	historical_python/search_snapshot.json; no live database in this archive
Low-genus candidates in implemented screens	0	Same checkpoint; not a nonexistence theorem
Class-size lower bound	74,173,379,224,089,213,153,040	Exact quadratic-form subgroup certificate

The Hess value is a necessary bound within a specified construction, not an actual genus-992 descent. The subgroup certificate is an unconditional lower bound. Equality with the PARI class number, and whole-class coverage by the selected generator set, rely on GRH-dependent class-group completeness.

The earlier degree-3/7 search preserves quadratic genus characters and does not reach the whole class. The later worker uses degrees 3,17,31 and canonicalizes all 155 absolute-Frobenius conjugates. It records direct GHS ranks for all proper subfields and the Hess necessary bound over F32. Screens are invariant on the conjugacy orbit; they are not 155 independent tests. See the historical source for exact candidate thresholds.



0 candidates in implemented screens. Not a full-class or security conclusion.

The figure displays counts copied from the checkpoint, with no sampling interval or attack-cost interpretation. A necessary genus bound is a filter, not a constructed descent. PDF report (reports/REPORT.pdf).

Evidence and verification status

historical_python/ retains the prepared source, compact JSON outputs, modular-polynomial inputs, exact GP certificate, and verification receipts. evidence-manifest.json binds every historical file by byte count and SHA256. The operational SQLite database, worker log, downloaded papers, and extracted PARI binary are not included. The aggregate search count is therefore a checkpoint observation; individual records from all 24,492 orbits cannot be independently replayed from this archive alone.

Historical verification receipts report field checks, the independent binary product-span exclusion, degree-three identities and point samples, SIGKILL recovery, deterministic resume comparison, file-lock exclusion, 18 modular-edge witnesses, and the exact GP subgroup certificate. These receipts are not an independent replay of the entire long-running search. No Lean formalization or identity.certificate/v1 promotion is claimed. The general class-group torsor implication is a literature theorem; the GP calculation is an arithmetic certificate, not a Lean proof of it.

Native continuation and prior work

The repository already has native direct-GHS arithmetic and a bounded census. Start with the existing Oakley study (../ghs_oakley_extended_20261023/REPORT.md), src/bin/ghs_screen.rs, examples/ghs_oakley_isogeny_census.rs, and src/cryptanalysis/binary_velu.rs. Reuse those implementations when replacing the historical degree-3/17/31 persistent worker. Its native replacement and independent replay have not been completed by this archival PR.

The previous Oakley report agrees on the seed's large direct descent to F32, the n=5 tower's genus 16, and the absence of a security conclusion. This archive adds historical support-screen and class-group evidence; it does not supersede that study's native measurements or extend its cost claims.

Sources reviewed during the original investigation and repository import:

- RFC 2409, section 6.3 (https://www.rfc-editor.org/rfc/rfc2409.html#section-6.3): exact field, model and order. - Jacobson, Menezes and Stein (2001) (https://www.math.uwaterloo.ca/~ajmeneze/publications/124.pdf): classical GHS parameters and comparison example. - Hess (2004) (https://doi.org/10.1112/S146115700000108X): generalized two-factor construction; only its specified support relaxation is tested. - Achter and Howe (2017), section 4 (https://msp.org/ant/2017/11-1/ant-v11-n1-p.pdf): ordinary isogeny-class torsors. - PARI 2.15 manual (https://pari.math.u-bordeaux.fr/pub/pari/manuals/2.15.5/users.pdf): exact form operations and conditional class-number completeness. - Sutherland modular-polynomial data (https://math.mit.edu/~drew/ClassicalModPolys.html): bundled input data. Attribution: Broeker, Lauter and Sutherland, *Modular polynomials via isogeny volcanoes*, Mathematics of Computation 81 (2012), 1201–1231.

Requirement status and workflow limits

Requirement	Status	Remaining gap
Preserve completed work in a reviewable PR	Implemented	GitHub review and applicable checks

Requirement	Status	Remaining gap
Search the entire class	Partial	Only the recorded bounded population has been screened
Find a low-genus representative	Not established	Zero candidates in the implemented screens
Resume natively in this repository	Open	Replace persistent worker using native arithmetic and storage
Independent whole-search replay	Open	Full operational database and a separate native replay are absent
Formal proof of general exclusions	Open	No universal exclusion or Lean proof is claimed

The diagram and PDF are an archival rendering of recorded values, not a new experiment or benchmark. No IC/rho ratio, phase cost, new exponent, scalar-recovery result, or best-of result is promoted. Accordingly the existing Oakley diagram, docs/index-calculus-scoreboard.html, docs/ic/progress-timeline.json and docs/ic-leaderboard.html are left unchanged; none of their compared costs is revised. A progress-metrics entry tracks the imported checkpoint population, explicitly as historical data.

Conductor could not be used in this execution environment: the command is not installed and .conductor/WORKFLOW.md returned HTTP 404 at the inspected base commit. Cargo and Lean are also unavailable here. No new cryptographic tests, full Rust-library build, EKS job, S3 checkpoint, or proof build was claimed for this import. Native continuation must satisfy those execution and replay requirements before its results are promoted.